

Assessment of cell death

Material:

1. Spinal cord segment
2. Ethidium homodimer-1
3. Carbogenic mixture
4. ACSF
5. Glucose 10mM

Procedures:

1. To analyze cell death, stain 400- μ m slices with 3 μ L of ethidium homodimer-1 (6 μ M; live/dead assay, Molecular Probes, Eugene, OR) diluted in 1 mL of carbogenic mixture for 30 min at room temperature in the dark.
2. Wash with 2 mL of 95 % O₂/5 % CO₂ ACSF containing glucose 10 mM at room temperature for 15 min. During the staining procedure, protect the slices from light.
3. To quantify the dead cells, their nuclei are identified by fluorescence staining with ethidium homodimer. Acquire the images with a Zeiss Axiovert 200 M Microscope and the Apotome system, which afford consecutive 10- μ m thick optical sections of a Z-series. These sections are analyzed at the ethidium homodimer excitation/emission wavelength (568/598 nm), with 20x dry objectives.
4. Use the Image J software to combine consecutive optical sections from a Z-series, to create image constructions, and to make a quantitative analysis. Count all the nuclei in this predetermined area.